SG No. 221 O_{h}^{1} $Pm\bar{3}m$ [cubic]

* plus set: +[0, 0, 0]

Table 1: Wyckoff site: 1a, site symmetry: m-3m

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 2: Wyckoff site: 1b, site symmetry: m-3m

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 3: Wyckoff site: 3c, site symmetry: 4/mm.m

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,17,18,19,20,25,26,27,28,41,42,43,44]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,6,7,8,13,14,15,16,29,30,31,32,37,38,39,40]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[9,10,11,12,21,22,23,24,33,34,35,36,45,46,47,48]

Table 4: Wyckoff site: 3d, site symmetry: 4/mm.m

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,2,3,4,17,18,19,20,25,26,27,28,41,42,43,44]
2	$\left[0, \frac{1}{2}, 0\right]$	[5,6,7,8,13,14,15,16,29,30,31,32,37,38,39,40]
3	$\left[0, 0, \frac{1}{2}\right]$	[9,10,11,12,21,22,23,24,33,34,35,36,45,46,47,48]

Table 5: Wyckoff site: 6e, site symmetry: 4m.m

No.	position	mapping
1	$[x, 0, 0]$	[1,4,17,20,26,27,42,43]
2	$[-x, 0, 0]$	[2,3,18,19,25,28,41,44]
3	$[0, x, 0]$	[5,8,13,16,30,31,38,39]
4	$[0, -x, 0]$	[6,7,14,15,29,32,37,40]
5	$[0, 0, x]$	[9,12,22,23,34,35,45,48]
6	$[0, 0, -x]$	[10,11,21,24,33,36,46,47]

Table 6: Wyckoff site: 6f, site symmetry: 4m.m

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1,4,17,20,26,27,42,43]
2	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,18,19,25,28,41,44]

continued ...

Table 6

No.	position	mapping
3	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[5,8,13,16,30,31,38,39]
4	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[6,7,14,15,29,32,37,40]
5	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[9,12,22,23,34,35,45,48]
6	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[10,11,21,24,33,36,46,47]

Table 7: Wyckoff site: 8g, site symmetry: .3m

No.	position	mapping
1	$[x, x, x]$	[1,5,9,38,43,48]
2	$[-x, -x, x]$	[2,7,12,37,41,45]
3	$[-x, x, -x]$	[3,8,10,39,44,46]
4	$[x, -x, -x]$	[4,6,11,40,42,47]
5	$[x, x, -x]$	[13,17,21,26,31,36]
6	$[-x, -x, -x]$	[14,19,24,25,29,33]
7	$[x, -x, x]$	[15,20,22,27,32,34]
8	$[-x, x, x]$	[16,18,23,28,30,35]

Table 8: Wyckoff site: 12h, site symmetry: mm2..

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,4,26,27]
2	$\left[-x, \frac{1}{2}, 0\right]$	[2,3,25,28]
3	$\left[0, x, \frac{1}{2}\right]$	[5,8,30,31]
4	$\left[0, -x, \frac{1}{2}\right]$	[6,7,29,32]
5	$\left[\frac{1}{2}, 0, x\right]$	[9,12,34,35]
6	$\left[\frac{1}{2}, 0, -x\right]$	[10,11,33,36]
7	$\left[\frac{1}{2}, x, 0\right]$	[13,16,38,39]
8	$\left[\frac{1}{2}, -x, 0\right]$	[14,15,37,40]
9	$\left[x, 0, \frac{1}{2}\right]$	[17,20,42,43]
10	$\left[-x, 0, \frac{1}{2}\right]$	[18,19,41,44]
11	$\left[0, \frac{1}{2}, -x\right]$	[21,24,46,47]
12	$\left[0, \frac{1}{2}, x\right]$	[22,23,45,48]

Table 9: Wyckoff site: 12i, site symmetry: m.m2

No.	position	mapping
1	$[0, y, y]$	[1,18,28,43]
2	$[0, -y, y]$	[2,20,27,41]
3	$[0, y, -y]$	[3,17,26,44]
4	$[0, -y, -y]$	[4,19,25,42]
5	$[y, 0, y]$	[5,15,32,38]
6	$[y, 0, -y]$	[6,13,31,40]
7	$[-y, 0, y]$	[7,16,30,37]

continued ...

Table 9

No.	position	mapping
8	$[-y,\ 0,\ -y]$	`[8,14,29,39]`
9	$[y,\ y,\ 0]$	`[9,21,36,48]`
10	$[-y,\ y,\ 0]$	`[10,23,35,46]`
11	$[y,\ -y,\ 0]$	`[11,22,34,47]`
12	$[-y,\ -y,\ 0]$	`[12,24,33,45]`

Table 10: Wyckoff site: `12j`, site symmetry: `m.m2`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ y\right]$	`[1,18,28,43]`
2	$\left[\frac{1}{2},\ -y,\ y\right]$	`[2,20,27,41]`
3	$\left[\frac{1}{2},\ y,\ -y\right]$	`[3,17,26,44]`
4	$\left[\frac{1}{2},\ -y,\ -y\right]$	`[4,19,25,42]`
5	$\left[y,\ \frac{1}{2},\ y\right]$	`[5,15,32,38]`
6	$\left[y,\ \frac{1}{2},\ -y\right]$	`[6,13,31,40]`
7	$\left[-y,\ \frac{1}{2},\ y\right]$	`[7,16,30,37]`
8	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[8,14,29,39]`
9	$\left[y,\ y,\ \frac{1}{2}\right]$	`[9,21,36,48]`
10	$\left[-y,\ y,\ \frac{1}{2}\right]$	`[10,23,35,46]`
11	$\left[y,\ -y,\ \frac{1}{2}\right]$	`[11,22,34,47]`
12	$\left[-y,\ -y,\ \frac{1}{2}\right]$	`[12,24,33,45]`

Table 11: Wyckoff site: `24k`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,28]`
2	$[0,\ -y,\ z]$	`[2,27]`
3	$[0,\ y,\ -z]$	`[3,26]`
4	$[0,\ -y,\ -z]$	`[4,25]`
5	$[z,\ 0,\ y]$	`[5,32]`
6	$[z,\ 0,\ -y]$	`[6,31]`
7	$[-z,\ 0,\ y]$	`[7,30]`
8	$[-z,\ 0,\ -y]$	`[8,29]`
9	$[y,\ z,\ 0]$	`[9,36]`
10	$[-y,\ z,\ 0]$	`[10,35]`
11	$[y,\ -z,\ 0]$	`[11,34]`
12	$[-y,\ -z,\ 0]$	`[12,33]`
13	$[y,\ 0,\ -z]$	`[13,40]`
14	$[-y,\ 0,\ -z]$	`[14,39]`
15	$[y,\ 0,\ z]$	`[15,38]`
16	$[-y,\ 0,\ z]$	`[16,37]`
17	$[0,\ z,\ -y]$	`[17,44]`
18	$[0,\ z,\ y]$	`[18,43]`
19	$[0,\ -z,\ -y]$	`[19,42]`

continued ...

Table 11

No.	position	mapping
20	$[0, -z, y]$	`[20,41]`
21	$[z, y, 0]$	`[21,48]`
22	$[z, -y, 0]$	`[22,47]`
23	$[-z, y, 0]$	`[23,46]`
24	$[-z, -y, 0]$	`[24,45]`

Table 12: Wyckoff site: `24l`, site symmetry: `m..`

No.	position	mapping
1	$\left[\frac{1}{2}, y, z\right]$	`[1,28]`
2	$\left[\frac{1}{2}, -y, z\right]$	`[2,27]`
3	$\left[\frac{1}{2}, y, -z\right]$	`[3,26]`
4	$\left[\frac{1}{2}, -y, -z\right]$	`[4,25]`
5	$\left[z, \frac{1}{2}, y\right]$	`[5,32]`
6	$\left[z, \frac{1}{2}, -y\right]$	`[6,31]`
7	$\left[-z, \frac{1}{2}, y\right]$	`[7,30]`
8	$\left[-z, \frac{1}{2}, -y\right]$	`[8,29]`
9	$\left[y, z, \frac{1}{2}\right]$	`[9,36]`
10	$\left[-y, z, \frac{1}{2}\right]$	`[10,35]`
11	$\left[y, -z, \frac{1}{2}\right]$	`[11,34]`
12	$\left[-y, -z, \frac{1}{2}\right]$	`[12,33]`
13	$\left[y, \frac{1}{2}, -z\right]$	`[13,40]`
14	$\left[-y, \frac{1}{2}, -z\right]$	`[14,39]`
15	$\left[y, \frac{1}{2}, z\right]$	`[15,38]`
16	$\left[-y, \frac{1}{2}, z\right]$	`[16,37]`
17	$\left[\frac{1}{2}, z, -y\right]$	`[17,44]`
18	$\left[\frac{1}{2}, z, y\right]$	`[18,43]`
19	$\left[\frac{1}{2}, -z, -y\right]$	`[19,42]`
20	$\left[\frac{1}{2}, -z, y\right]$	`[20,41]`
21	$\left[z, y, \frac{1}{2}\right]$	`[21,48]`
22	$\left[z, -y, \frac{1}{2}\right]$	`[22,47]`
23	$\left[-z, y, \frac{1}{2}\right]$	`[23,46]`
24	$\left[-z, -y, \frac{1}{2}\right]$	`[24,45]`

Table 13: Wyckoff site: `24m`, site symmetry: `..m`

No.	position	mapping
1	$[x, x, z]$	`[1,38]`
2	$[-x, -x, z]$	`[2,37]`
3	$[-x, x, -z]$	`[3,39]`
4	$[x, -x, -z]$	`[4,40]`
5	$[z, x, x]$	`[5,48]`
6	$[z, -x, -x]$	`[6,47]`
7	$[-z, -x, x]$	`[7,45]`

continued ...

Table 13

No.	position	mapping
8	$[-z, x, -x]$	[8,46]
9	$[x, z, x]$	[9,43]
10	$[-x, z, -x]$	[10,44]
11	$[x, -z, -x]$	[11,42]
12	$[-x, -z, x]$	[12,41]
13	$[x, x, -z]$	[13,26]
14	$[-x, -x, -z]$	[14,25]
15	$[x, -x, z]$	[15,27]
16	$[-x, x, z]$	[16,28]
17	$[x, z, -x]$	[17,36]
18	$[-x, z, x]$	[18,35]
19	$[-x, -z, -x]$	[19,33]
20	$[x, -z, x]$	[20,34]
21	$[z, x, -x]$	[21,31]
22	$[z, -x, x]$	[22,32]
23	$[-z, x, x]$	[23,30]
24	$[-z, -x, -x]$	[24,29]

Table 14: Wyckoff site: 48n, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-x, -y, z]$	[2]
3	$[-x, y, -z]$	[3]
4	$[x, -y, -z]$	[4]
5	$[z, x, y]$	[5]
6	$[z, -x, -y]$	[6]
7	$[-z, -x, y]$	[7]
8	$[-z, x, -y]$	[8]
9	$[y, z, x]$	[9]
10	$[-y, z, -x]$	[10]
11	$[y, -z, -x]$	[11]
12	$[-y, -z, x]$	[12]
13	$[y, x, -z]$	[13]
14	$[-y, -x, -z]$	[14]
15	$[y, -x, z]$	[15]
16	$[-y, x, z]$	[16]
17	$[x, z, -y]$	[17]
18	$[-x, z, y]$	[18]
19	$[-x, -z, -y]$	[19]
20	$[x, -z, y]$	[20]
21	$[z, y, -x]$	[21]
22	$[z, -y, x]$	[22]
23	$[-z, y, x]$	[23]
24	$[-z, -y, -x]$	[24]

continued ...

Table 14

No.	position	mapping
25	$[-x, -y, -z]$	[25]
26	$[x, y, -z]$	[26]
27	$[x, -y, z]$	[27]
28	$[-x, y, z]$	[28]
29	$[-z, -x, -y]$	[29]
30	$[-z, x, y]$	[30]
31	$[z, x, -y]$	[31]
32	$[z, -x, y]$	[32]
33	$[-y, -z, -x]$	[33]
34	$[y, -z, x]$	[34]
35	$[-y, z, x]$	[35]
36	$[y, z, -x]$	[36]
37	$[-y, -x, z]$	[37]
38	$[y, x, z]$	[38]
39	$[-y, x, -z]$	[39]
40	$[y, -x, -z]$	[40]
41	$[-x, -z, y]$	[41]
42	$[x, -z, -y]$	[42]
43	$[x, z, y]$	[43]
44	$[-x, z, -y]$	[44]
45	$[-z, -y, x]$	[45]
46	$[-z, y, -x]$	[46]
47	$[z, -y, -x]$	[47]
48	$[z, y, x]$	[48]